MEDIUM · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Geometry and Trigonometry

01

10 answers · Rationale-rich · Teach from doc

Q1

B

B. 60°

Choice B is correct. It’s given that angle ACE measures $62°$. Since segments AE and BD are parallel, angles BDC and CEA are congruent. Therefore, angle CEA measures $58°$. The sum of the measures of angles ACE, CEA, and CAE is $180°$ since the sum of the interior angles of triangle ACE is equal to $180°$. Let the measure of angle CAE be $x°$. Therefore, $62 + 58 + x = 180$, which simplifies to $x = 60$. Thus, the measure of angle CAE is $60°$.

Choice A is incorrect. This is the measure of angle AEC, not that of angle CAE. Choice C is incorrect. This is the measure of angle ACE, not that of CAE. Choice D is incorrect. This is the sum of the measures of angles ACE and CEA.

Q2

D

D. 24

Choice D is correct. The perimeter, P, of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of 12 centimeters. Substituting 12 for s in the formula for the perimeter of a square yields $P = 412$, or $P = 48$. Therefore, the perimeter of square X is 48 centimeters. It’s also given that the perimeter of square Y is 2 times the perimeter of square X. Therefore, the perimeter of square Y is 248, or 96, centimeters. Substituting 96 for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by 4 gives $24 = s$. Therefore, the length of one side of square Y is 24 centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

13

The correct answer is 13. The equation of a circle in the xy-plane can be written in the form $(x-h)^2+(y-k)^2=r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy-plane is defined by $(x+2)^2+(y+5)^2=169$. Therefore, $r^2=169$. Taking the positive square root of both sides of this equation yields $r=13$. Thus, the radius of the circle is 13.

Q4

45

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 15. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are $45°$. It follows that the value of x is 45.

Q5

D

D. $100°$

Choice D is correct. The sum of the three interior angles in a triangle is $180°$. It's given that angle A measures $50°$. If angle B measured $100°$, the measure of angle C would be $180° - (50° + 100°) = 30°$. Thus, the measures of the angles in the triangle would be $50°$, $100°$, and $30°$. However, an isosceles triangle has two angles of equal measure. Therefore, angle B can't measure $100°$.

Choice A is incorrect. If angle B has measure $50°$, then angle C would measure $180° - (50° + 50°) = 80°$, and $50°$, $50°$, and $80°$ could be the angle measures of an isosceles triangle. Choice B is incorrect. If angle B has measure $65°$, then angle C would measure $180° - (65° + 50°) = 65°$, and $50°$, $65°$, and $65°$ could be the angle measures of an isosceles triangle. Choice C is incorrect. If angle B has measure $80°$, then angle C would measure $180° - (80° + 50°) = 50°$, and $50°$, $80°$, and $50°$ could be the angle measures of an isosceles triangle.

Q6

A

A. 1,764

Choice A is correct. The area, A, of a square is given by the equation $A = s^2$, where s is the length of one side of the square. Let x represent the length, in feet, of one side of square N. It's given that the area of square N is 49 square feet; therefore, it follows that $49 = x^2$. It's also given that the length of one side of square M is 6 times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression $6x$. Substituting $6x$ for s in the equation $A = s^2$ yields $A = (6x)^2$, or $A = 36x^2$. Since $49 = x^2$, substituting 49 for x^2 in the equation $A = 36x^2$ yields $A = 36(49)$, or $A = 1{,}764$. Therefore, the area, in square feet, of square M is $1{,}764$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

B

B. $\frac{1}{3}\pi$

Choice B is correct. The measure of an angle, in radians, can be found by multiplying its measure, in degrees, by $\frac{\pi}{180}$. It's given that the measure of angle Z is $60°$. It follows that the measure, in radians, of angle Z is $60\frac{\pi}{180}$, or $\frac{1}{3}\pi$.

Choice A is incorrect. This is the measure, in radians, of an angle whose measure is $30°$, not $60°$.

Choice C is incorrect. This is the measure, in radians, of an angle whose measure is $120°$, not $60°$.

Choice D is incorrect. This is the measure, in radians, of an angle whose measure is $180°$, not $60°$.

Q8

45

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 32. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are 45°. It follows that the value of x is 45.

Q9

B

B. $x > 115$

Choice B is correct. In the figure shown, the angle measuring $y°$ is congruent to its vertical angle formed by lines s and m, so the measure of the vertical angle is also $y°$. The vertical angle forms a same-side interior angle pair with the angle measuring $x°$. It's given that lines r and s are parallel. Therefore, same-side interior angles in the figure are supplementary, which means the sum of the measure of the vertical angle and the measure of the angle measuring $x°$ is $180°$, or $x + y = 180$. Subtracting x from both sides of this equation yields $y = 180 - x$. Substituting $180 - x$ for y in the inequality $y < 65$ yields $180 - x < 65$. Adding x to both sides of this inequality yields $180 < 65 + x$. Subtracting 65 from both sides of this inequality yields $115 < x$, or $x > 115$. Thus, if $y < 65$, it must be true that $x > 115$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $x + y$ must be equal to, not less than, 180.

Choice D is incorrect. $x + y$ must be equal to, not greater than, 180.

Q10 B

B. 350

Choice B is correct. The area, A, of a triangle is given by $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is the corresponding height of the triangle. It's given that a triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. Substituting 10 for b and 70 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}1070$, or $A = 350$. Therefore, the area, in square centimeters, of the triangle is 350.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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